💼 CFO AI Assistant

Intelligent Financial Analysis for Nigerian Banks

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📋 Project Overview

An AI-powered assistant designed for Nigerian bank CFOs to streamline financial analysis, policy queries, and market intelligence gathering through natural language conversations.

15 Nigerian Banks	**3** AI Agents
4 Policy Documents	**<5s** Response Time

🎯 Problem Statement

❌ Manual Data Analysis

CFOs spend hours querying databases, running Excel formulas, and creating reports manually.

❌ Document Search

Searching through 50+ page policy documents to find specific approval thresholds or procedures.

❌ Delayed Decisions

Waiting for analysts to compile competitive intelligence and market insights.

✅ Solution

- ✓ **Natural Language Queries:** Ask questions in plain English, get instant answers
- ✓ **Multi-Agent Architecture:** Specialized agents for SQL, RAG, and Web Search
- ✓ **Real Banking Data:** 15 Nigerian banks with comprehensive financial metrics
- ✓ **Policy Intelligence:** Instant access to approval processes and compliance requirements
- ✓ **Conversation Memory:** Context-aware follow-up questions

✨ Key Features

📊 SQL Analysis

Text-to-SQL Queries
Convert natural language to SQL queries. Analyze financial metrics, ratios, and trends across Nigerian banks.

📚 RAG Search

Policy Document Retrieval
Semantic search through policy documents using vector embeddings. Find relevant policies instantly.

"Show banks with ROE above 20%"

"What's the approval for ₦50M loans?"

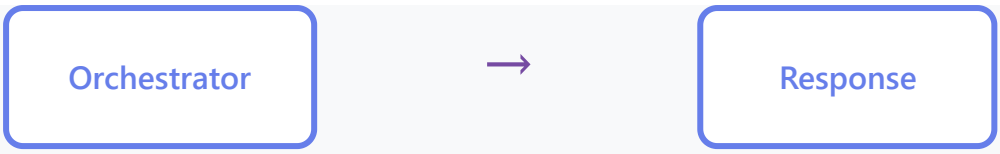
🌐 Web Intelligence

Real-time Market Data
Search the web for current market trends, CBN policies, and regulatory updates.

"Latest CBN monetary policy changes"

🏗️ System Architecture





🛠️ Technology Stack

Claude Sonnet 4 (Anthropic) | LangChain | ChromaDB | SQLite | Streamlit | Sentence Transformers | Python 3.9+ | Tavily API

Component Breakdown

🤖 LLM

Claude Sonnet 4 - Advanced reasoning, code generation, and natural language understanding

🗄️ Vector Database

ChromaDB - Efficient similarity search for document retrieval

🔗 Orchestration

LangChain - Agent coordination, memory management, and chain composition

💻 Interface

Streamlit - Interactive web UI with real-time updates

💾 Database Schema

Nigerian Banking Data

```
companies: - company_id (INT) - bank_name (TEXT) - ticker (TEXT) -
sector (TEXT) - market_cap_millions (REAL) financial_statements: -
company_id (INT) - year (INT) - quarter (TEXT) -
total_revenue_millions (REAL) - net_income_millions (REAL) -
total_assets_millions (REAL) - shareholders_equity_millions (REAL)
financial_ratios: - company_id (INT) - roe (REAL) - roa (REAL) -
capital_adequacy_ratio (REAL) - npl_ratio (REAL) -
loan_to_deposit_ratio (REAL)
```

Banking Metrics Included

- ✓ **Profitability:** ROE, ROA, Net Profit Margin, Gross Margin
- ✓ **Capital:** Capital Adequacy Ratio (CAR), Equity Ratios
- ✓ **Asset Quality:** NPL Ratio, Loan-to-Deposit Ratio
- ✓ **Efficiency:** Cost-to-Income Ratio, Operating Efficiency
- ✓ **Liquidity:** Liquidity Ratio, Cash Reserves

💬 Sample Queries

SQL Queries

"Show top 5 banks by total assets"

"Compare ROE across Tier 1 banks"

"Which banks have CAR above 15%?"

"Average NPL ratio in banking sector"

Policy Queries

"What's the approval for ₦100M loans?"

"CBN capital adequacy requirements"

"Investment policy restrictions"

"Foreign exchange management rules"

Web Queries

"Current CBN monetary policy rate"

"Recent Nigerian banking regulations"

"Digital banking trends in Nigeria"

"Inflation impact on banks"

⚙ Implementation Highlights

1. Query Routing Logic

Intelligent routing determines which agent handles each query:

```
if query contains ["show", "compare", "rank", "which"]: route to SQL Agent elif query contains ["policy", "approval", "requirement"]: route to RAG Agent else: route to Web Agent
```

2. RAG Pipeline

- ✓ Documents chunked using LangChain's RecursiveCharacterTextSplitter
- ✓ Embeddings generated with Sentence Transformers (all-MiniLM-L6-v2)
- ✓ Vector similarity search in ChromaDB
- ✓ Context-aware responses with source citations
- ✓ Conversation memory for follow-up questions

3. Text-to-SQL

- ✓ Schema-aware SQL generation

- ✓ Automatic query validation
- ✓ Results formatted with natural language explanations
- ✓ Visual charts for numeric data

🚀 Deployment Options

🎈 Streamlit Cloud

Best for: Quick demos
✓ Free hosting
✓ Easy deployment
✓ GitHub integration

🤗 Hugging Face Spaces

Best for: ML projects
✓ Free tier
✓ GPU support
✓ Community visibility

☁️ Cloud Platforms

Best for: Production
✓ Scalable
✓ Custom domains
✓ Enterprise features

🔮 Future Enhancements

📈 Historical Analysis

Time-series data (2020-2024) for trend analysis and forecasting

🔐 Authentication

User authentication and role-based access control

📊 Advanced Viz

🔚 Live Data

Interactive dashboards with drill-down capabilities

Real-time connection to banking APIs and data warehouses

🤝 Collaboration

Shared workspaces and report generation

📱 Mobile App

Native iOS/Android applications

📬 Contact & Links

Project Information

Built by: Micheal Olaniyi
Role: AI Engineer
Focus: Nigerian Banking + AI

GitHub: github.com/Purepen/CFO-ai-assistant/
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